

Name: Pragati Gupta
Class: AIDS - B
PRN: 12413775
Roll no: 51 Batch: B2

TUTORIAL: 1

- Q Study and Understand the Foundational Concept, Working, Mechanism, Advantages and Limitation of Bitcoins.

→ Introduction: Bitcoin is the world's first decentralized digital cryptocurrency. It was introduced in 2008 by an anonymous person or group using the pseudonym Satoshi Nakamoto & became operational in January 2009. Bitcoin uses blockchain technology, a distributed ledger that records all transactions securely without requiring a central authority such as a bank or government.

Foundational Concept:

- ① Decentralization: Bitcoin operates on peer-to-peer network where no single organization controls the system. Every participant maintains a copy of the blockchain.
- ② Blockchain: A blockchain is a distributed digital ledger that stores transaction records in blocks. Each block is linked to the previous one using cryptographic hashes, making the data highly secure & tamper-resistant.
- ③ Cryptography: Bitcoin uses cryptographic techniques such as:
 - a) SHA-256 Hashing for securing blocks.
 - b) Public and Private Key Cryptography for ownership & digital signatures.

- ④ Mining: Mining is the process of validating transactions and adding new blocks to the blockchain. Miners solve complex mathematical puzzles and receive Bitcoin as a reward.
- ⑤ Consensus Mechanism: Bitcoin uses the Proof of Work consensus algorithm. It ensures that all nodes agree on the same version of the blockchain.

Component:

- ① Blockchain: A distributed ledger that stores all Bitcoin transactions in chronological order.
- ② Blocks: A block contains
- Transaction data
 - Timestamp
 - Previous block hash
 - Current block hash
 - Nonce
- ③ Nodes: Computers connected to the Bitcoin network that validate and relay transactions.
- ④ Miners: Special nodes that verify transactions and create new blocks using the Proof of Work consensus mechanism.
- ⑤ Wallets: Applications or devices used to store Bitcoin private and public keys.

Working Mechanism:

Step 1: Transaction Creation

A user creates a transaction by specifying the recipient's Bitcoin address and the amount to transfer.

Step 2: Digital Signature

The transaction is digitally signed up using the sender's private key to verify ownership.

Step 3: Broadcasting

The signed transaction is broadcast to the Bitcoin peer to peer network.

Step 4: Verification

Network nodes verify the transaction by checking:

- Valid digital signature
- Availability of sufficient balance
- Transaction validity.

Step 5: Mining

Verified transactions are grouped into a block. Miners compete to solve a cryptographic puzzle using the Proof of Work algorithm.

Step 6: Block Addition

The first miner to solve the puzzle adds the new block to the blockchain.

Step 7: Confirmation

Once the block is added, the transaction receives confirmation and becomes permanent and immutable.

Advantages:

- ① Decentralized system with no central authority.
- ② High level of security with using cryptography.
- ③ Transparent transaction history through blockchain.
- ④ Fast international money transfers.
- ⑤ Lower transaction fees compared to traditional banking.
- ⑥ Protection against double spending.
- ⑦ Limited supply of 21 millions Bitcoins help control inflation.
- ⑧ Users have complete control over their digital assets.

Limitation:

- ① High price volatility.
- ② Large energy consumption due to Proof of work mining.
- ③ Limited transaction processing speed.
- ④ Scalability challenges during high network traffic.
- ⑤ Regulatory uncertainty in many countries.
- ⑥ Transaction cannot be reversed once confirmed.
- ⑦ Loss of private keys results in permanent loss of Bitcoins.
- ⑧ Can be misused for illegal activities due to pseudonymity.

Application:

- ① Online payments
- ② International remittances
- ③ Digital investment asset
- ④ E-commerce transactions
- ⑤ Store of value

Conclusion:

Bitcoin is first successful implementation of blockchain technology and has transformed the concept of digital finance. Its decentralized architecture, cryptographic security and transparent ledger make it one of the most innovative financial technologies. Although Bitcoin has challenges such as scalability, energy consumption and price volatility, it has paved the way for cryptographic and numerous blockchain based applications.